Master Motivation Letter

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Recruitment Panel
Barcelona Supercomputing Center - Centro Nacional de Supercomputación
Barcelona, Spain

Subject: Application for AI, machine-learning, and scientific software engineering opportunities

Dear Members of the Recruitment Panel,

I am writing to express my interest in research and engineering opportunities at the Barcelona Supercomputing Center involving agentic AI, generative AI applications, machine learning for scientific systems, and reliable software infrastructure. I am completing an MSc in Modelling for Science and Engineering at the Universitat Autonoma de Barcelona, specialising in Data Science, after a BSc in Mechanical Engineering completed in Budapest under a Stipendium Hungaricum scholarship. This path has given me an interdisciplinary profile across physical modelling, software engineering, data systems, and language-model-based applications.

At Aily Labs, my work has followed the progression from prompt-defined MCP capabilities to gated skills, semantic data access, and production evaluation. For analytical workflows, a model needs more than a database connection. It needs approved table descriptions, business meaning, and usage guidance so an SQL agent can select and query data through a bounded interface. Package-owned question sets then declare expected tables, tools, skills, and answer criteria, while deterministic checks, traces, and LLM-as-a-judge scoring provide an Agent IQ view of capability. This creates a practical improvement loop spanning MCP and Super Agent services, local integration, CI/CD, semantic discrepancies, and promotion checks. It has taught me that model quality cannot be separated from the harness that supplies context, permissions, tools, and evidence.

In parallel, my JAE Intro ICU 2025 scholarship at IIIA-CSIC has focused on a modular interactive architecture for the NAO robot. I have worked across ROS 2 components for dialogue, grounded symbolic state, structured planning, deterministic action dispatch, execution feedback, and replanning. The resulting MSc thesis evaluates the integrated system through controlled dialogue, knowledge-base interaction, simple and composite skills, and staged failure management. This work made the harness problem concrete: generated decisions must remain inspectable, constrained by reviewed capabilities, and tied to execution evidence before success is reported.

My earlier engineering experience also shapes how I approach AI systems. At ABS Consulting, I supported seismic qualification and structural assessment for UK nuclear power assets. Working with Abaqus, Mathcad, Inventor, STAAD.Pro, technical standards, and client deliverables reinforced habits of traceability, careful validation, and clear technical communication. My BSc thesis applied a neural network to turbulent-intensity prediction from wind-profile data, connecting machine learning with fluid and physical modelling.

Outside formal work, iTrader is my principal machine-learning research project. Its implemented RLVR-style baseline combines a task-conditioned PPO/GAE solver, FiLM policy and value networks, validated task specifications, decomposed rewards for PnL, drawdown and turnover, local LLM proposer backends, and reproducible run manifests. Its intended destination is live trading, but only after capability-profiled solver heads, grouped proposer optimisation, Market-JEPA representations, replay and shadow evaluation, and deterministic risk and execution gates have been demonstrated.

I have also built an unpublished H0 proof for a Universal Agentic Harness. Its abstraction-boundary framework treats the LLM as a role-bounded policy inside a larger coupled system and compiles the capabilities, context, permissions, and evidence paths needed for each task. The Watson Stack explores local model serving and tool use through LiteLLM, llama.cpp, and a Hermes-style agent loop. These projects are not published or externally validated, but they represent the question connecting my work: how to improve model capability through the surrounding system while preserving evidence, evaluation, and deterministic control.

BSC is especially attractive to me because it brings AI research, high-performance computing, and scientific application domains into the same institution. I would contribute strong Python and systems thinking, experience working across model, tool, data, and execution boundaries, and a willingness to learn the deeper HPC and post-training methods required by the team. I am comfortable working independently, but I value multidisciplinary collaboration and precise communication. Having studied and worked in Hungary, the United Kingdom, and Spain, I am also at ease in international environments and work fluently in English and Spanish.

I would welcome the opportunity to discuss where this combination of engineering discipline, agentic AI development, and scientific modelling could contribute most effectively at BSC.

Yours sincerely,

Juan David Bendek Williamson

References

1. **Roger Montane**, Mid Data Scientist, Aily Labs

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2. **Dr. Balogh Miklos**, Professor, BME

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